

PORTFOLIO · 2026

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I take ideas to shipped software,
alone, with AI agents — and set the
rules that keep them from breaking.

UBC Sauder, Class of 2026 · Finance + Business Analytics

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SIX MONTHS, ONE PERSON

Four kinds of things I built

The numbers are output, not users, as of Oct 2026.

45

projects

1,389

logged updates

26

apps on one private database

33

public repos

Delivered to others

- Letterkeep
Reference-letter system · in use by a professor
- Query Mirror
SQL teaching explorer · no-code tool for a professor

Public

- Scholar Outflow Lab
13 countries, recomputed weekly
- CallDelta
Earnings-call sentiment · tone → residual signal
- Vacancy Signal
Ghost job postings as an equity signal · 310K postings

Used every day

- 26-app personal operating system

Runs on its own

- Multi-agent content pipeline
topic to release pack
- 10 scheduled jobs
- Guardrails and sentinels
machine blocks dangerous operations

What I built, and the decision inside each

WORK 1 · DELIVERED TO A PROFESSOR

Letterkeep

Reference-letter system

Roster in, students screened by grade and attendance, documents uploaded, emails out. Shipped as a web app in a day; rebuilt as a desktop app after reading the privacy law, so student data stays on the professor's machine.

3 months · 56 iterations · Python / Flask / SQLite / OCR

THE POINT Nobody asked. I read the law and changed the architecture.

WORK 2 · PUBLIC

nickkk-skills

21 open-source agent skills for Claude Code

Nine make a deliverable: a one-page data report, a three-statement financial model, a slide deck. Twelve check or guard an agent's work: an evidence audit, self-tests that must be able to fail, a git guardrail hook, a privacy gate before publishing.

MIT · 20 of 21 tested in Claude Code and in Codex · [NickkkLian/nickkk-skills](#)

THE POINT "Done, tested" is a claim until someone can check it.

WORK 3 · RUNS ON ITS OWN

Multi-agent content pipeline

Topic to release pack across several content channels. Agents exchange files, never chat; resumable state machine, bounded retries, a QC loop that sends failures back, and a human always publishes.

32 modules · 20 skills · usage ledger reconciled to the platform bill

THE POINT My own ledger covered 1.2% of the real bill. The bill became the source of truth.

WORK 4 · KNOWLEDGE

Structured knowledge bases built with AI

Thousands of entries across several private libraries. AI does the entry work; every entry carries a permanent ID and a source; reuse is by pointer, never by copy.

Verbatim first · no lossy simplification · checks before and after every change

THE POINT The hard part is the acceptance discipline, not the AI.

Six of ten public projects

Each one is a complete, documented tool. Also public: Letterkeep, Headersort, Callback Desk, Bill Bench, and the 21 agent skills in nickkk-skills.

Query Mirror

No-code SQL teaching tool for a professor: click tables, filters and groups, the SQL appears live. Deterministic by design, no LLM.

Vanilla JS · sql.js (SQLite in WebAssembly) vendored
[NickkkLian/SQL-Viz-for-Edu](#)

Scholar Outflow Lab

Where researchers from 13 countries end up, from OpenAlex open data. Ranked by Wilson lower bound so small samples sink. Recomputed weekly.

Python stdlib scraper · GitHub Pages · launchd
[NickkkLian/Scholar-Outflow-Lab](#)

CallDelta

Management framing vs analyst pushback, by topic, across quarters. Does tone explain the residual move net of the EPS surprise?

Python · JSON schema + Pydantic · static dashboard
[NickkkLian/Earnings-Call-Sentiment-Analyser](#)

Vacancy Signal

Ghost postings in 310K listings, scored per firm. Backtest 2019–2026: the long/short lost 22.1%; both tails long roughly matched SPY. Team of three.

XGBoost · GMM · SHAP · Revelio data
[NickkkLian/Ghost-Job-Detection-And-Trading-Signal](#)

Clipbind

Saved Xiaohongshu posts and Bilibili videos in one library; Claude merges selected notes into one writeup. Data stays in your browser and your own repo.

Single-page app · IndexedDB · MIT · live demo
[NickkkLian/content-organizer](#)

ApplyLedger

A multi-region job-search tracker in one HTML file: prompt generators, batch resume tailoring, and the GitHub Contents API as its backend.

React 18 in one HTML file · GitHub Contents API
[NickkkLian/Job-Tracker](#)

Ship first, then let machines keep it honest

Most guardrail rules come from a real incident. I don't hand-write code: I define the problem, set the constraints, verify the result, and own the result.

Keeping systems reliable

- A pre-tool hook blocks dangerous operations
force push, mass deletion, pushing while behind · 421 lines · ask by default
- Sentinels raise alarms
database shrinks, copies drift, sessions collide
- The reviewer is never the doer
an audit agent judges evidence only
- Three backup tiers, restores actually rehearsed

What I can do

- Take an idea to production alone
- Set the rules that keep systems reliable
- Python backends, web frontends, agent orchestration
AI sessions in parallel, one repo each
- Product judgment: what is worth building, what to cut
- Business grounding: valuation, financials, backtesting
bank valuation work · CFA Level I